# REFINED VALUATIVE CRITERION FOR PROPERNESS ALONG LIMITS

THE STACKS PROJECT AUTHORS

*Editorial note.* The reduction to the affine valuation ring case is the standard locality ('immediate'); the constructive content is fully worked in the counted span.

*Allowed standard prerequisites (editorial).*

- Properness of limits. Limits, properness lemmas of the chapter
- Valuative criterion. Morphisms, valuative criterion lemmas

*Source reference.* Original reference 04II: sources/stacks-chow/tags.txt, source lines 5904–5904. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 07RM: sources/stacks-chow/tags.txt, source lines 9872–9872. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01RA: sources/stacks-chow/tags.txt, source lines 2295–2295. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

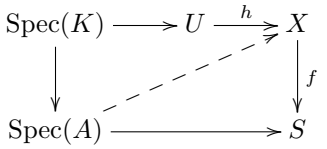
*Source reference.* Original lemma 0208: sources/stacks-chow/tags.txt, source lines 2625–2625. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01KZ: sources/stacks-chow/tags.txt, source lines 2068–2068. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0BX5: sources/stacks-chow/tags.txt, source lines 14968–14968. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $f : X \to S$ and $h : U \to X$ be morphisms of schemes. Assume that $S$ is locally Noetherian, that $f$ and $h$ are of finite type, that $f$ is separated, and that $h(U)$ is dense in $X$. If given any commutative solid diagram*




*Date*: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*where $A$ is a discrete valuation ring with field of fractions $K$, there exists a dotted arrow making the diagram commute, then $f$ is proper.*

*Proof.* There is an immediate reduction to the case where $S$ is affine. Then $U$ is quasi-compact. Let $U = U_1 \cup \ldots \cup U_n$ be an affine open covering. We may replace $U$ by $U_1 \amalg \ldots \amalg U_n$ without changing the assumptions, hence we may assume $U$ is affine. Thus we can find an open immersion $U \to Y$ over $X$ with $Y$ proper over $X$. (First put $U$ inside $\mathbf{A}^n_X$ using Morphisms, Lemma 04II and then take the closure inside $\mathbf{P}^n_X$, or you can directly use Morphisms, Lemma 07RM.) We can assume $U$ is dense in $Y$ (replace $Y$ by the scheme theoretic closure of $U$ if necessary, see Morphisms, Section 01RA). Note that $g : Y \to X$ is surjective as the image is closed and contains the dense subset $h(U)$. We will show that $Y \to S$ is proper. This will imply that $X \to S$ is proper by Morphisms, Lemma 03GN thereby finishing the proof. To show that $Y \to S$ is proper we will use part (4) of Lemma 0208. To do this consider a diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{y} & Y \\ \downarrow & \nearrow & \downarrow{\scriptstyle f\circ g} \\ \mathrm{Spec}(A) & \longrightarrow & S \end{array}$$

where $A$ is a discrete valuation ring with fraction field $K$ and where $y : \mathrm{Spec}(K) \to Y$ is the inclusion of a generic point. We have to show there exists a unique dotted arrow. Uniqueness holds by the converse to the valuative criterion for separatedness (Schemes, Lemma 01KZ) since $Y \to S$ is separated as the composition of the separated morphisms $Y \to X$ and $X \to S$ (Schemes, Lemma 01KU). Existence can be seen as follows. As $y$ is a generic point of $Y$, it is contained in $U$. By assumption of the lemma there exists a morphism $a : \mathrm{Spec}(A) \to X$ such that

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \xrightarrow{y} & U & \longrightarrow & X \\ \downarrow & \nearrow_{a} & & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & & \longrightarrow & & S \end{array}$$

is commutative. Then since $Y \to X$ is proper, we can apply the valuative criterion for properness (Morphisms, Lemma 0BX5) to find a morphism $b : \mathrm{Spec}(A) \to Y$ such that

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{y} & Y \\ \downarrow & \nearrow_{b} & \downarrow{\scriptstyle g} \\ \mathrm{Spec}(A) & \xrightarrow{a} & X \end{array}$$

is commutative. This finishes the proof since $b$ can serve as the dotted arrow above. □